

Pre-Registered Experimental Protocol

Skill sweeps, threshold ablations, distribution shift, and held-out evaluation

Core Research Document - Next computational experiment

Human-AI Codecision Systems Research Program

Stage One revision - September 2026

Evidence status: Synthetic Stage One contains five independent seeds, 180 trials per policy and seed, 3,600 total decisions, and a controlled AI distribution shift in the final quarter. There were zero human participants. Adaptive codecision selectively regulated authority but did not improve synthetic accuracy or declared utility over human-only control. Human trust, satisfaction, and real cognitive state remain unmeasured.

1. Rationale

Synthetic Stage One established selective authority regulation but not outcome superiority. The next confirmatory computational experiment therefore tests whether this null/negative result is caused by threshold choice, human/AI skill relationships, or policy structure.

2. Confirmatory hypotheses

ID	Hypothesis
H0-1	No pre-specified adaptive threshold configuration improves declared utility over human-only and static controls across held-out seeds.
H1-1	At least one pre-specified adaptive configuration reproducibly improves declared utility without exceeding the unnecessary-override constraint.
H0-2	Relative human/AI skill does not materially interact with intervention benefit.
H1-2	Intervention benefit depends materially on relative human/AI skill and/or distribution shift.

3. Experimental controls

- Keep event generation, trial budgets, and seed policy matched across policy comparisons.
- Separate parameter-discovery seeds from confirmatory held-out seeds.
- Preserve human-only, static recommendation, adaptive codecision, and aggressive automation controls.
- Pre-specify the unnecessary-override and autonomy constraints used to define acceptable utility.
- Retain all valid seeds, including poor adaptive-policy outcomes.
- Record distribution-shift onset and severity in the manifest.

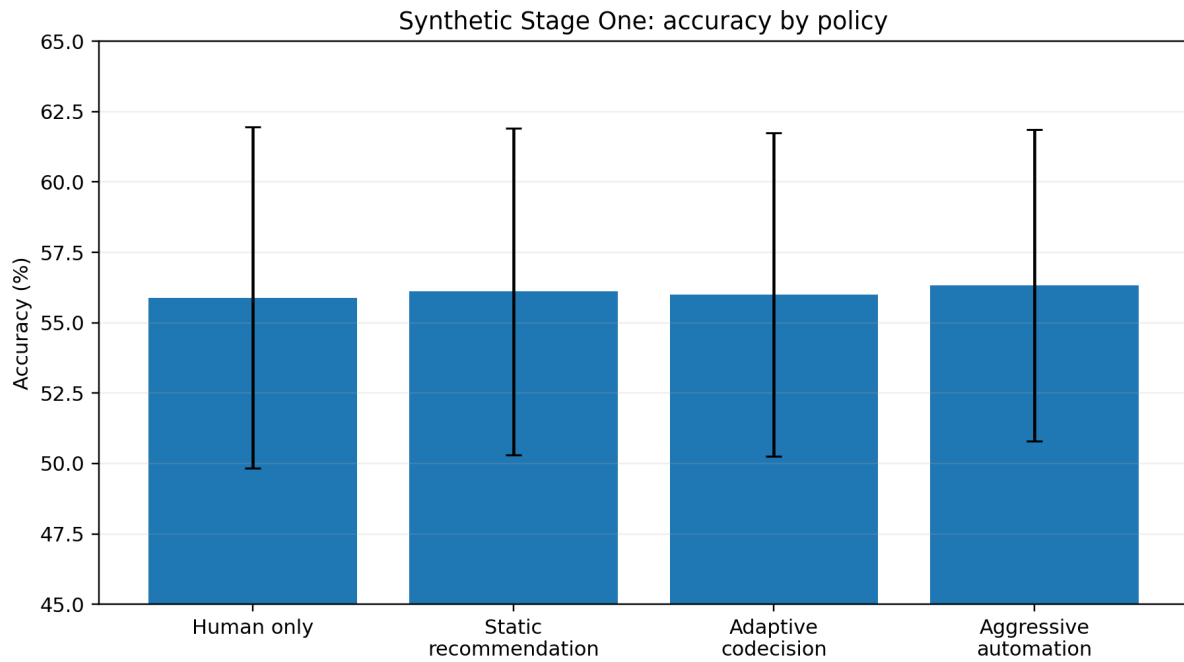


Figure 1. Frozen Stage One accuracy reference; not part of the future confirmatory dataset.

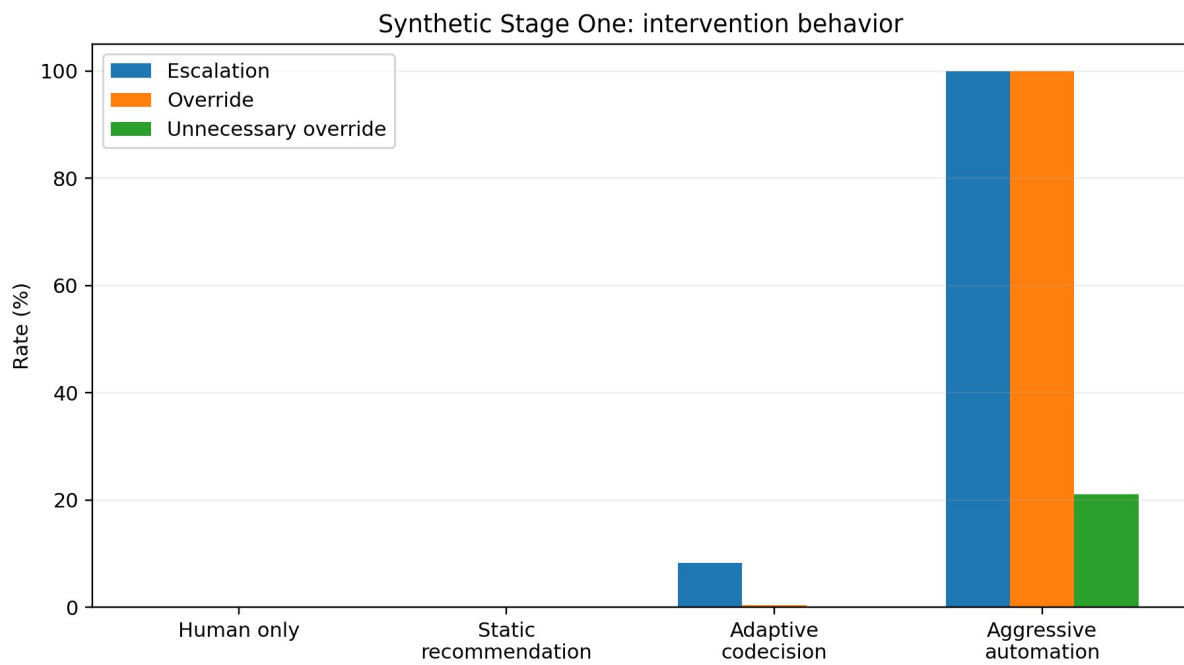


Figure 2. Frozen Stage One intervention reference.

4. Statistical plan

Use seed-level replication for condition comparisons, report uncertainty intervals and effect sizes, and control multiplicity across large threshold grids. Threshold discovery and final confirmation

must use different seed sets. A favorable configuration discovered after inspecting confirmatory results must be labeled exploratory and re-tested.

5. Decision rule

Adaptive codecision will be described as outcome-improving only if it reproducibly improves the pre-specified utility/performance criterion while satisfying autonomy and unnecessary-override constraints. Otherwise the supported conclusion remains selective authority regulation without demonstrated outcome superiority.